

SIH 26143 Deep Research Report — Satellite Oil-Spill Detection, Hindcasting & Vessel Attribution

Executive Summary

Problem Statement 26143 — NTRO ka real challenge sirf satellite image mein oil detect karna nahi hai. Actual challenge ek **maritime forensic pipeline** banana hai:

Oil slick detect karo → real oil vs look-alike decide karo → spill ka probable origin aur time reverse-track karo → us space-time window mein vessels reconstruct karo → possible source vessels ko rank karo → uncertainty aur evidence ke saath explainable report do.

Ye distinction important hai. Sentinel-1 par oil segmentation relatively mature research problem hai; difficult aur high-value part hai **false-positive reduction + physical hindcasting + AIS correlation + responsible-vessel attribution without falsely accusing a ship**. Existing operational systems—especially EMSA CleanSeaNet aur KSAT—already SAR oil detection aur AIS/context fusion karte hain; CleanSeaNet oil-drift forecasting/backtracking bhi provide karta hai. Isliye winning novelty “SAR + AIS ko combine kiya” nahi ho sakti. ¹

Meri recommended SIH architecture:

Sentinel-1 SAR → Oil Segmentation → Look-alike Rejection → Ensemble Hindcast → Probabilistic Origin Envelope → AIS Trajectory Reconstruction → Explainable Vessel Ranking → “Unknown Source” hypothesis → Evidence/Provenance Report

Is architecture ka strongest differentiator hoga **uncertainty-aware attribution**. System ko “Vessel X culprit hai” nahi bolna chahiye. Output hona chahiye:

Vessel X — investigative likelihood score 0.78, 90% uncertainty interval 0.61–0.87; closest approach 1.3 km; temporal offset 11 min; trajectory compatibility high; AIS continuity good.

Aur simultaneously:

Unknown / non-AIS source — score 0.14

IMO guidance khud warn karti hai ki AIS complete traffic picture nahi deta, kuch vessels AIS carry nahi karte, AIS switch off ho sakta hai, aur AIS ko sole information source nahi maana jaana chahiye. Isliye “unknown source” option technically aur legally bahut important hai. ²

Recommended primary data stack:

- **Detection:** Sentinel-1 GRD VV/VH via Copernicus Data Space Ecosystem.
- **Cross-check:** Sentinel-2 L2A; Landsat 8/9; MODIS only broad contextual monitoring.
- **Operational drift:** Copernicus Marine **SMOC** hourly surface currents because it explicitly combines ocean circulation + tides + Stokes drift; Copernicus itself recommends this product for drift/Lagrangian applications. ³
- **Wind:** ECMWF Open Data for operational forecasts; ERA5 for historical back-testing.
- **Trajectory engine:** OpenDrift/OpenOil as main implementation; NOAA GNOME/WebGNOME as independent validation engine.
- **AIS demo:** MarineCadastre historical data / synthetic AIS; global production integration via MarineTraffic or another licensed provider.
- **ML:** U-Net as baseline and SegFormer as stronger candidate; DeepLabv3+ as a useful second CNN baseline. ⁴

Most important MVP scope decision: 8 weeks mein “everything” mat build karna. First working chain should be:

one Sentinel-1 scene → reliable mask → one hindcast ensemble → AIS candidates → ranked suspects + confidence → map + PDF evidence report.

Age estimation, spoof detection, optical fusion, multi-model uncertainty aur dark-vessel satellite detection ko advanced phase mein rakho.

From a CEO/CFO/product-risk perspective, **PS 26143 is a GO**, but only if team project ko “AI image classifier” nahi, **decision-support and forensic evidence platform** ki tarah build kare.

Real-World Pain, Stakeholders and Operational Constraints

Problem ground reality mein kahan difficult hai

SAR mein oil slick often relatively dark appear karta hai because surface oil small capillary waves ko damp karta hai, reducing radar backscatter. Problem ye hai ki **har dark region oil nahi hota**. Low-wind areas, natural films, oceanographic features aur other phenomena oil-like dark signatures produce kar sakte hain. Recent Sentinel-1 benchmark work explicitly oil slicks ke saath large “look-alike” class include karta hai because robust systems ko oil detect karne ke saath false detections avoid bhi karna hota hai. ⁵

PS ka second difficulty temporal hai: satellite ek slick ko **spill ke baad** observe karta hai. Current, wind, waves aur turbulence usko source se shift kar chuke hote hain. Isliye image mein nearest ship ko culprit bolna scientifically wrong ho sakta hai. Copernicus Marine ka dedicated surface merged-current product general circulation ke saath tides aur wave Stokes drift combine karta hai specifically surface-drift applications ke liye. ³

Third difficulty AIS hai. AIS observations irregular ho sakte hain, receiver coverage incomplete ho sakti hai, messages absent ho sakte hain, aur AIS data alone complete operational picture nahi deta. IMO specifically cautions against depending on AIS as the sole source of navigation information. ²

Stakeholders aur unko exactly kya chahiye

Stakeholder	Real decision	Unko system se kya chahiye	Wrong output ka damage
NTRO / intelligence analyst	Incident ko quickly characterize aur source hypotheses narrow karna	Automated imagery analysis, historical reconstruction, vessel ranking, explainability	Wrong vessel focus, analyst time waste, intelligence credibility loss
Coast Guard / maritime enforcement	Kahan response bhejna hai? Kis vessel ko inspect karna hai?	Position, spill extent, drift forecast, candidate tracks, confidence	Response delay ya innocent vessel investigation
Environmental agency	Spill kidhar jayega aur vulnerable coast/ resources ko kya risk hai?	Forward forecast, area, uncertainty envelope, repeat observations	Cleanup resources wrong location par deploy ho sakte hain
Port authority	Suspect vessel next port par inspect/hold/ review karna hai?	MMSI/IMO, route, ETA, evidence timeline	Operational disruption / weak case
Insurer / P&I / claims teams	Liability aur event chronology establish karna	Reproducible source data, time stamps, attribution confidence, extent	Wrong liability allocation / claims disputes
Investigator / legal authority	Model output evidence-worthy supporting material hai ya sirf lead?	Provenance, original product IDs, chain of processing, uncertainty, analyst audit	Evidence challenged or over-interpreted

Operational platforms already show ki **speed matters**. CleanSeaNet analysed satellite imagery national authorities ko near-real-time deliver karta hai; KSAT says its analysed oil-spill reports typically satellite acquisition ke 20–120 minutes ke andar deliver hote hain. ¹

Lekin SIH architecture mein latency ko do separate numbers mein measure karo:

$$T_{\text{alert}} = T_{\text{satellite availability}} + T_{\text{download}} + T_{\text{preprocess}} + T_{\text{inference}} + T_{\text{hindcast}} + T_{\text{AIS}} + T_{\text{report}}$$

Team ka control mainly last six components par hai; satellite acquisition timing tum control nahi kar sakte.

False positive is not a minor metric

Agar oil detector 95% recall deta hai but har SAR scene mein dozens of false oil slicks generate karta hai, operationally system weak hai. Jury ke saamne sirf Dice/IoU nahi dikhana.

Measure:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$IoU = \frac{TP}{TP + FP + FN}$$

Saath mein report karo:

- False alarms per scene
- False alarms per 1000 km^2
- Oil vs look-alike confusion
- Detection confidence calibration
- Minimum detectable slick area

Recent Eastern Mediterranean benchmark specifically oil objects aur look-alike/other patches separate karta hai, making it useful for exactly this validation. ⁵

Legal/evidentiary constraint

Satellite + AIS attribution **investigative evidence** ban sakta hai, but model score ko conviction-style conclusion banana risky hai. EMSA documentation shows satellite-derived pollution information and AIS/ vessel information operational enforcement mein use hote hain, and documented European cases illustrate that imagery should be combined with supporting information rather than treated as isolated proof. ⁶

Therefore report wording:

Bad:

“Ship ABC caused the spill.”

Good:

“Ship ABC is the highest-ranked investigative candidate under the stated satellite, AIS and hindcast assumptions.”

For every case preserve:

```
raw product ID → source URI → acquisition timestamp → SHA-256 → preprocessing
config → model version/hash → forcing dataset/version → simulation parameters →
AIS snapshot hash → score calculation → analyst actions → exported report
```

Ye proposed engineering chain-of-custody hai; legal admissibility jurisdiction-specific counsel ko determine karni hogi.

Existing Solution Landscape

Sabse important research finding: **NTRO ka problem completely new category nahi hai**. Several mature systems individual pieces solve karte hain, aur CleanSeaNet almost complete operational reference architecture ke closest hai.

Solution	Provider	SAR / optical detection	Hindcast / forecast	AIS / vessel correlation	Uncertainty	Data	Cost / licence
CleanSeaNet	EMSA	SAR yes; optical on request	Yes , oil drift forecasting/ backtracking	Yes , AIS + vessel detections + ancillary info	Detection confidence; full probabilistic attribution UQ not publicly specified	Sentinel/ commercial SAR, met/ oceanographic, AIS	Govern service particip states
Oil Spill Detection Service	KSAT	SAR primary	Not explicitly offered on cited public product page	Yes	Human QA; probabilistic UQ not publicly described	Multi-constellation SAR + AIS	Comm quote
Maritime Insights / RADARSAT maritime monitoring	MDA Space	SAR oil + ship detection	Not publicly specified in cited material	Vessel monitoring; exact AIS-attribution workflow not publicly specified	Not public	RADARSAT-2 + geospatial context	Comm
OSCAR	SINTEF	No built-in satellite detector	Yes, hindcast + forecast	No native AIS attribution advertised	Stochastic/ statistical modelling supported	Met/ocean forcing + oil properties	Comm contact provide
WebGNOME / GNOME	NOAA	No	Yes	No	Yes — ensemble/ uncertainty concepts	User/met-ocean forcing	Public/ NOAA
OpenDrift / OpenOil	MET Norway / community	No	Yes, forward + backward	No	Yes, stochastic/ ensemble capable	OPeNDAP/ THREDDS/ current/wind/ waves	GPL op source

Solution	Provider	SAR / optical detection	Hindcast / forecast	AIS / vessel correlation	Uncertainty	Data	Cost / licence
DARTIS-related Sentinel-1 system	Research collaboration	Sentinel-1 SAR	Research-system dependent	No full vessel-attribution pipeline demonstrated in cited detection work	Research-level	Sentinel-1	Academic

Sources and product details: CleanSeaNet officially combines SAR analysis with meteorological, oceanographic and AIS/vessel information and provides oil-drift forecasting/backtracking in its interface.

⁷ KSAT uses multiple SAR constellations, algorithmic processing, human quality control and AIS correlation, reporting typical delivery 20–120 minutes after acquisition. ⁸ MDA describes RADARSAT-2-supported near-real-time ship/oil surveillance and maritime monitoring services. ⁹ SINTEF OSCAR supports hindcasting, forecasting and stochastic modelling of marine oil releases. ¹⁰ OpenDrift is an open-source Lagrangian framework supporting backwards simulation and stochastic uncertainty. ¹¹

Official links:

- EMSA CleanSeaNet: <https://www.emsa.europa.eu/csn-menu/csn-service/122/2360.html>
- KSAT Oil Spill Detection: <https://www.ksat.no/earth-observation/environmental-monitoring/oil-spill-detection-service/>
- MDA Space: <https://mda.space/>
- SINTEF OSCAR: <https://www.sintef.no/en/software/oscar/>
- NOAA GNOME: <https://response.restoration.noaa.gov/oil-and-chemical-spills/oil-spills/response-tools/gnome.html>
- OpenDrift: <https://opendrift.github.io/>
- OpenDrift source: <https://github.com/OpenDrift/opendrift>

Competition se actual novelty kya honi chahiye

Do not pitch:

“Existing systems detect oil, hum AI + AIS combine kar rahe hain.”

CleanSeaNet aur KSAT already woh story invalidate kar denge. ¹

Pitch instead:

“We are building an auditable probabilistic maritime-forensics pipeline that propagates uncertainty from satellite segmentation through physical hindcasting into AIS-based source ranking, while explicitly retaining an unknown-source hypothesis.”

Strong novelty components:

1. **Uncertainty propagation end-to-end**, not only detection confidence.
2. **Unknown/non-AIS vessel hypothesis**, reducing forced false attribution.
3. **Explainable score decomposition** — spatial, temporal, heading, overlap, behavioural and data-quality evidence separately.
4. **Model-independent evidence bundle** with hashes and versions.
5. **Multi-modal false-positive rejection** using SAR texture/polarization + wind + temporal data + optional optical evidence.
6. **Two independent drift engines during validation** — OpenDrift main, GNOME cross-check.
7. **India-oriented architecture** where AIS provider can be replaced without retraining CV/drift pipeline.
8. **Human-in-the-loop final disposition**, similar to the quality-control principle used operationally by KSAT. ⁸

Data Inventory and Training Benchmarks

Operational satellite and maritime data

Important 2026 correction: old **Copernicus Open Access Hub / SciHub is retired**. Its operations ended in late 2023; new projects should use **Copernicus Data Space Ecosystem (CDSE)** via STAC, OData, S3 or Sentinel Hub. ¹²

Source	Direct endpoint / API	Format & cadence	Access limits / cost	Best use in SIH26143
Sentinel-1 SAR — CDSE	STAC: https://stac.dataspace.copernicus.eu/v1/ OData: https://catalogue.dataspace.copernicus.eu/odata/v1/Products	GRD/SLC, SAFE; COG-SAFE GRD available; mission acquisition driven	Free account; fair-use/ service quotas	Primary oil detector
Sentinel-2 — CDSE	Same STAC/OData; Sentinel Hub Process: https://sh.dataspace.copernicus.eu/process/v1	L1C/L2A; optical multispectral	Free/fair-use; clouds matter	Optical confirmation and look-alike check
Old Copernicus Open Access Hub	https://scihub.copernicus.eu/	Legacy	Retired	Do not build new code against it
AWS Sentinel-2 L2A	s3://sentinel-s2-l2a/	Sentinel-2 L2A	Public AWS dataset	Cloud workflow / bulk training

Source	Direct endpoint / API	Format & cadence	Access limits / cost	Best use in SIH26143
AWS Sentinel-2 COG Collection	s3://e84-earth-search-sentinel-data/	Cloud-optimized GeoTIFF	Open AWS dataset	Fast tile access
Landsat 8/9	STAC: https://landsatlook.usgs.gov/stac-server M2M: https://m2m.cr.usgs.gov/	COG/STAC/USGS products; archive continuously updated	USGS open data	Historical optical confirmation
Landsat AWS	s3://usgs-landsat/collection02/	Collection 2	Requester Pays	Large-scale back-testing
MODIS Terra/ Aqua	CMR: https://cmr.earthdata.nasa.gov/search/ LAADS: https://ladsweb.modaps.eosdis.nasa.gov/	HDF; global coverage ~1–2 days, NRT products within hours	NASA Earthdata account for many downloads	Large spill/context; not fine attribution
MarineCadastre AIS	https://marinecadastre.gov/accessais/ Bulk: https://marinecadastre.gov/ais/	Historical US AIS, GIS-ready records	Free; current AccessAIS ordering service is reported unavailable, bulk remains	SIH demo/ history , not India production
AISHub	https://data.aishub.net/ws.php	JSON/XML/CSV near-real-time	API for contributors; minimum request interval ~1 minute	Low-cost real-time development where coverage exists
MarineTraffic / Kpler	Docs: https://servicedocs.marinetraffic.com/ Base pattern: https://services.marinetraffic.com/api/...	Commercial AIS API	Paid/contract; API limits contract dependent, published hard ceiling up to 100 requests/min	Production global AIS

Source	Direct endpoint / API	Format & cadence	Access limits / cost	Best use in SIH26143
NOAA CoastWatch ERDDAP	https://coastwatch.noaa.gov/erddap/	NetCDF/CSV/JSON via ERDDAP depending dataset	Public	Secondary ocean observations/ validation
HYCOM ESPC-D-V02	Catalog: https://tds.hycom.org/thredds/catalog.html	NetCDF/ OPeNDAP; 1/12° global analysis, active 2024-present, forecast	Free; demonstration service, typical availability within ~48h	Backup current model / research
OSCAR NRT v2	DOI: https://doi.org/10.5067/OSCAR-25N20 CMR collection search available via Earthdata	NetCDF; global 0.25°, daily, NRT latency ~2 days	NASA Earthdata	Smooth large-scale current/ backtest
Copernicus Marine SMOC	Product: GLOBAL_ANALYSISFORECAST_PHY_001_024 ; dataset cmems_mod_glo_phy_anfc_merged-uv_PT1H-i	Hourly, 1/12°, NetCDF , daily updates, forecast	Free registration	Recommended primary drift forcing
ECMWF Open Data winds	https://data.ecmwf.int/forecasts/	GRIB2; current IFS/ AIFS operational forecasts	Open-data licence	Operational wind
ERA5	CDS: https://cds.climate.copernicus.eu/	NetCDF/ GRIB; hourly historical fields	Free account/ API; delayed relative to operational forecasts	Historical spill back-testing

CDSE officially exposes STAC, OData, S3 and Sentinel Hub access, including Sentinel-1 GRD/SLC and Sentinel-2 products. ¹³ Sentinel-1 COG_SAFE GRD products are also available to simplify cloud-native raster access. ¹⁴

The AWS Registry exposes Sentinel-2 L1C/L2A and COG collections; **do not assume there is one equivalent globally complete raw Sentinel-1 AWS public bucket**. For global S1, CDSE is the safer primary source; AWS Sentinel-1 offerings found in the registry are often region-specific derived products. ¹⁵

USGS exposes Landsat through its M2M and STAC interfaces; Collection 2 also has an AWS bucket. ¹⁶
NASA MODIS provides frequent global coverage and near-real-time products useful for large spills, although its coarse 250–1000 m native band resolutions make it secondary to Sentinel-1 for fine slick boundaries. ¹⁷

MarineCadastre data originates from the U.S. Coast Guard AIS network and is repurposed by NOAA/BOEM; it includes fields such as time, position, vessel type, speed and vessel dimensions. As of the currently indexed official page, its interactive ordering service is unavailable while bulk downloads remain available.

¹⁸ That means **MarineCadastre is excellent for algorithm development, but not a production Indian AIS feed.**

AISHub provides JSON/XML/CSV API access but its free real-time model is built around members contributing AIS receiver feeds, with an API request interval of at least one minute. ¹⁹ MarineTraffic/Kpler APIs are commercial and support current/historical vessel information; limits and historical-access depth depend on contract. ²⁰

Which ocean-current source should we actually use?

For the prototype, my ordering is:

Primary: Copernicus Marine SMOC

Secondary: HYCOM

Back-test / sanity: OSCAR

Independent observational context: NOAA ERDDAP

Copernicus SMOC is particularly relevant because its hourly surface product explicitly includes:

$$u_{\text{total}} = u_{\text{ocean}} + u_{\text{tide}} + u_{\text{Stokes}}$$

$$v_{\text{total}} = v_{\text{ocean}} + v_{\text{tide}} + v_{\text{Stokes}}$$

at 1/12° grid resolution, and Copernicus documentation recommends it for drift/Lagrangian applications.

³

HYCOM's active ESPC-D-V02 provides 1/12° global outputs, but the HYCOM service says its daily data can typically become accessible within about 48 hours, so it is better as an alternate forcing source than as your only low-latency operational dependency. ²¹

OSCAR v2 provides daily 0.25° surface-current estimates, with final/interim/NRT quality levels and about 2-day latency for NRT. It is useful for broad-scale backtesting rather than high-resolution coastal incident response. ²²

Training and benchmarking datasets

Dataset	What it contains	Use	Direct link
Sentinel-1 SAR Oil Spill Dataset Part I	1,200 oil-spill VV/VH Sigma0 images, 2048×2048×2 + masks	Training/validation segmentation	https://zenodo.org/records/8346860
Part II	Companion part of same three-part collection	Non-oil / complementary training	https://doi.org/10.5281/zenodo.8253899
Part III	Test set: 150 oil + 150 look-alike + 150 no-oil , masks	Strong held-out benchmark	https://zenodo.org/records/13761290
Eastern Mediterranean oil/look-alike benchmark	3,225 oil objects across 1,365 patches + 2,290 other/look-alike patches	False-positive and domain-shift test	https://doi.org/10.1594/PANGAEA.980773
Krestenitis et al. oil-spill dataset	Roughly ~1,000 Sentinel-1 images; oil/look-alike/land/ship/sea classes	Literature benchmark	Access by proposal rather than fully open
Sen1Floods11	4,831 Sentinel-1 flood/water chips	SAR encoder pretraining / augmentation only	https://github.com/cloudtostreet/Sen1Floods11

The Zenodo Part I dataset provides georeferenced dual-polarization VV/VH Sigma0 imagery and pixel masks under CC BY 4.0. ²³ Part III deliberately contains oil, look-alike and no-oil cases, which is exactly what a robust false-positive test needs. ²⁴

The newer Eastern Mediterranean public dataset provides a much stronger look-alike challenge and is linked to an openly described research pipeline. ⁵

Sen1Floods11 ko oil-spill benchmark mat bolna. It is a flood/surface-water Sentinel-1 dataset, not oil ground truth. It can help pretrain SAR feature representations or validate preprocessing, but final oil performance must be measured on actual oil-spill datasets. ²⁵

Dataset split strategy

Randomly adjacent patches ko train/test mein split karna dangerous hai, because same original SAR acquisition ke neighboring tiles information leak kar sakte hain.

Prefer:

scene-level split or **event/geography-level split**

Example:

- Training: Mediterranean + Mexico/other events
- Validation: completely unseen scenes
- Test A: Zenodo Part III
- Test B: Eastern Mediterranean look-alike dataset
- Test C: one manually reconstructed historical real spill case

Final jury numbers should include **cross-geography performance**, not only random-patch accuracy.

Algorithms and Proposed Multi-Modal Architecture

Oil slick segmentation

Three reasonable models:

U-Net should be baseline. Its encoder-decoder structure is simple, data-efficient and easy to debug. ²⁶

DeepLabv3+ is useful because atrous spatial pyramid pooling handles multi-scale context and its decoder refines boundaries. Oil slicks can be long/thin/irregular, so multi-scale context is relevant. ²⁷

SegFormer is my production candidate if GPU/resources permit because its hierarchical transformer extracts multi-scale representations with a lightweight decoder. ²⁸

Recommended experiment:

Model	Purpose
U-Net + ResNet encoder	Baseline
DeepLabv3+	CNN comparison
SegFormer-B1/B2	Candidate final model
Ensemble of best 2	Advanced uncertainty

Input channels:

$$X = [\sigma_{VV}^0, \sigma_{VH}^0, VV - VH, \text{local texture}, \text{incidence angle}]$$

Not every dataset provides every feature cleanly, so start with **VV + VH**, then add engineered channels as ablation experiments.

Loss:

$$L = \lambda_1 L_{\text{BCE}} + \lambda_2 L_{\text{Dice}} + \lambda_3 L_{\text{boundary}}$$

Dice handles class imbalance; boundary term can improve thin slick outlines.

False-positive reduction

This is probably where a winning solution differentiates.

For each detected polygon compute:

SAR appearance

- mean VV/VH backscatter
- contrast against neighboring ocean
- local standard deviation
- entropy
- GLCM-like texture
- edge roughness

Geometry

- area
- perimeter
- elongation
- compactness
- skeleton length
- tail-like structure
- orientation

Polarization

- VV
- VH
- VV/VH ratio or difference
- incidence-angle-normalized statistics

Environmental context

- wind speed
- current direction
- nearby ship tracks
- repeat-scene behaviour

Multi-temporal

- Did feature appear suddenly?
- Is it moving in physically plausible direction?
- Does it persist like a natural low-wind feature?

Optical evidence from Sentinel-2 can be used **when cloud-free**, but SAR remains primary because radar works through clouds and independent of daylight—a major reason operational systems such as CleanSeaNet and KSAT rely heavily on SAR. ¹

A useful architecture is:

$$P(\text{oil}) = f(P_{\text{segmentation}}, \text{texture}, \text{polarization}, \text{geometry}, \text{wind}, \text{temporal context})$$

The second-stage classifier could initially be XGBoost/LightGBM instead of another massive neural network. Jury ko feature importance dikhana easy hoga.

Hindcasting physics

Represent slick as particles.

Basic surface-particle motion:

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}_{\text{ocean}} + \mathbf{u}_{\text{tide}} + \mathbf{u}_{\text{Stokes}} + \beta \mathbf{u}_{\text{wind}} + \mathbf{u}_{\text{diffusion}}$$

where:

- $\mathbf{u}_{\text{ocean}}$ = ocean current,
- $\mathbf{u}_{\text{tide}}$ = tidal current,
- $\mathbf{u}_{\text{Stokes}}$ = wave-induced Stokes drift,
- $\beta \mathbf{u}_{\text{wind}}$ = wind-driven effect/windage parameterization,
- diffusion = unresolved turbulence.

OpenDrift is explicitly a Lagrangian trajectory framework capable of taking currents, winds and waves from multiple sources, running both forward and backward, and handling stochastic/ensemble uncertainty. 11

Forward:

$$\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + \mathbf{v}(\mathbf{x}_t, t) \Delta t$$

Hindcast:

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \mathbf{v}(\mathbf{x}_t, t) \Delta t$$

Actual numerical integration should use OpenDrift rather than a hand-built Euler equation for production.

Hindcast ko probabilistic banao

Ek single reverse line produce mat karo.

Uncertainty sources:

- segmentation boundary
- unknown spill age
- current uncertainty
- wind uncertainty
- turbulence

- oil/weathering properties
- forcing model choice

For every incident, generate e.g. hundreds or thousands of particles across plausible initial slick pixels and parameter perturbations.

Result:

50% probable origin region
90% probable origin region
probable release-time interval

NOAA GNOME explicitly uses trajectory uncertainty for spill-response planning; OpenDrift also supports stochastic and ensemble approaches. ²⁹

Conceptually:

$$p(\mathbf{x}_0, t_0 \mid \text{slick}) \approx \frac{1}{N} \sum_{i=1}^N K_h(\mathbf{x}_0 - \mathbf{x}_0^{(i)})$$

where N = hindcast ensemble members.

AIS trajectory reconstruction

Raw AIS first clean karo:

- duplicate timestamps
- impossible latitude/longitude
- impossible speed jumps
- MMSI validation
- gaps
- abrupt impossible displacement
- timestamp timezone normalization
- irregular-sampling interpolation

Each ship ko trajectory:

$$T_v = \{(lat_i, lon_i, t_i, SOG_i, COG_i)\}_{i=1}^n$$

mein convert karo.

Then origin envelope se relevant candidates retrieve:

$$d_v = \min_{t \in W} distance(T_v(t), O(t))$$

where $O(t)$ is probabilistic spill-origin region and W plausible spill-time window.

Vessel scoring model

Recommended features:

Group	Feature
Spatial	minimum origin distance
Spatial	fraction of AIS track inside 50/90% origin envelope
Temporal	minimum release-time offset
Direction	vessel heading vs inferred source-to-slick direction
Dynamics	speed during possible discharge
Behaviour	stop/slowdown/course change
Integrity	AIS gap near release window
Data quality	number/freshness of valid AIS observations
Vessel context	vessel type, size — weak prior only
Model robustness	rank consistency across hindcast ensembles

Spatial likelihood:

$$L_s(v) = \exp\left(-\frac{d_v^2}{2\sigma_x^2}\right)$$

Temporal likelihood:

$$L_t(v) = \exp\left(-\frac{\Delta t_v^2}{2\sigma_t^2}\right)$$

Heading agreement:

$$L_h(v) = \frac{1 + \cos(\Delta\theta_v)}{2}$$

Then:

$$z_v = w_1 \log L_s + w_2 \log L_t + w_3 L_h + w_4 O_v + w_5 B_v + w_6 Q_v - w_7 G_v$$

where:

- O_v = origin-envelope overlap
- B_v = behavioural compatibility
- Q_v = data-quality score
- G_v = uncertainty/gap penalty

Now **unknown-source hypothesis** add karo:

$$P(v | E) = \frac{e^{z_v}}{e^{z_{unknown}} + \sum_j e^{z_j}}$$

This is strategically very important.

Agar sab vessels poor match hain:

system should say **“No AIS-observed vessel can be attributed with sufficient confidence.”**

rather than forcibly selecting rank #1.

Also, until retrospective cases par calibration perform nahi hoti, UI mein raw model score ko “probability of guilt” mat call karo. Better label:

Investigative compatibility score

Uncertainty propagation

For every ensemble run r :

$$z_v^{(r)}$$

calculate karo.

Then output:

$$\text{Median}(z_v), \quad P_5(z_v), \quad P_{95}(z_v)$$

Example:

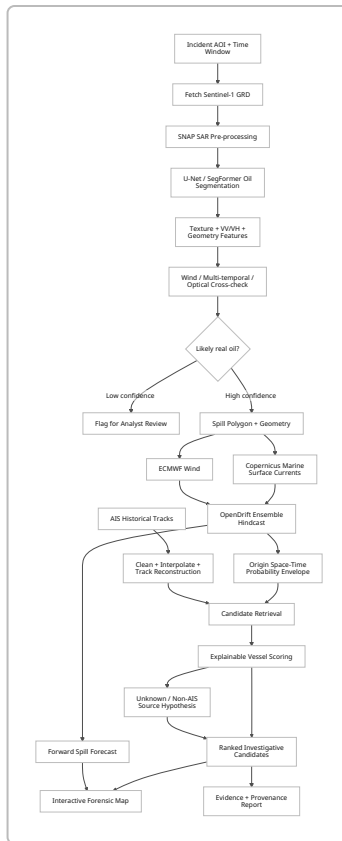
Candidate	Median score	90% interval
Vessel A	0.79	0.63–0.88
Vessel B	0.31	0.15–0.54
Unknown source	0.18	0.09–0.36

If rankings flip heavily across hindcast runs, analyst sees **low stability**.

Proposed system architecture

SIH26143 proposed architecture

[Download the architecture schematic](#)



Product Features, Tech Stack, APIs and Build Roadmap

MVP versus advanced version

Component	MVP — must work	Advanced — after MVP
Case creation	AOI + time + upload/select imagery	Automated surveillance queue
SAR	Sentinel-1 GRD ingest	Multi-sensor commercial SAR
Preprocessing	calibration, terrain correction, dB conversion	RTC/COG optimized pipeline
Detection	U-Net/SegFormer binary mask	Ensemble segmentation
False-positive	texture + confidence + wind	polarization + temporal + optical fusion
Spill geometry	area/length/orientation	approximate age interval
Drift	OpenDrift backward ensemble	Multi-model OpenDrift + GNOME
Currents	CMEMS SMOC	CMEMS + HYCOM + OSCAR ensemble
Wind	ECMWF	Multiple wind products

Component	MVP — must work	Advanced — after MVP
AIS	CSV/historical/synthetic ingest	Live paid API
Candidate selection	proximity/time	Bayesian/probabilistic score
AIS anomaly	basic gaps	spoofing/dark-vessel indicators
Unknown source	Yes — MVP	richer non-AIS model
UI	map + layers + ranked candidates	investigation workspace
Evidence	PDF + GeoJSON + JSON metadata	signed/auditable case bundle

UI/UX jury demo

Desktop screen:

Incident 043 Arabian Sea 14 Aug 2026 ANALYSIS COMPLETE		
CASE DATA	INTERACTIVE MAP	SUSPECTS
Sentinel-1 ✓		1. Vessel A
Wind ✓		Score 0.79
Currents ✓		2. Vessel B
AIS ✓		Score 0.31
Layers	Oil Slick	Unknown 0.18
☒ SAR	50% Origin	[Explain]
☒ Spill mask	90% Origin	
☒ AIS	— Vessel Tracks	
☒ Hindcast	→ Drift Direction	
02:00 — 03:00 — 04:00 — 05:00 FORENSIC TIMELINE		

Click **Explain**:

Spatial compatibility: 92/100
 Temporal compatibility: 86/100
 Heading compatibility: 73/100
 Origin overlap: 90/100
 AIS continuity: 98/100
 Behaviour anomaly: 40/100
 Ensemble rank stability: 88/100

Required jury outputs

One analysis should automatically produce:

- Original Sentinel-1 map
- Oil probability map
- Binary slick polygon
- oil/look-alike confidence
- area, length, centroid, orientation
- backward particle tracks
- 50% and 90% probable origin contours
- forward drift forecast
- reconstructed AIS traffic
- ranked candidate vessels
- candidate explanations
- uncertainty intervals
- timeline
- “unknown source” score
- PDF report
- GeoJSON/KML export
- machine-readable JSON
- data/model provenance manifest

Recommended technical stack

Layer	Recommendation
Core language	Python 3.11/3.12
SAR preprocessing	ESA SNAP + S1TBX , GDAL
Raster processing	Rasterio, GDAL, rioarray
Vector GIS	GeoPandas, Shapely, pyproj
ML	PyTorch
Segmentation	segmentation-models-pytorch / Hugging Face SegFormer
Scientific arrays	NumPy, xarray
Large datasets	Dask , optionally Zarr
Ocean modelling	OpenDrift/OpenOil
Independent drift validation	NOAA GNOME / PyGNOME
AIS	Pandas/Polars, GeoPandas, movingpandas/custom interpolation
API backend	FastAPI
Database	PostgreSQL + PostGIS

Layer	Recommendation
Queue	Celery/RQ + Redis
Object storage	S3 / MinIO
Frontend	React / Next.js
Map	MapLibre GL JS + deck.gl
Charts	Plotly / ECharts
Containers	Docker + Docker Compose
CI	GitHub Actions
Production-scale optional	Kubernetes

For SAR, ESA/CDSE preprocessing resources support Sentinel-1 GRD access, while COG_SAFE products can reduce some cloud I/O friction. ³⁰

Compute recommendation

For prototype:

Development/inference

- 8+ CPU cores
- 32 GB RAM
- 100–300 GB SSD
- NVIDIA GPU optional for serving

Training

- NVIDIA T4/L4/A10/RTX-class GPU
- preferably **16 GB+ VRAM**
- mixed precision
- crop/chip-based training instead of full 20k-pixel SAR scenes

You don't need multi-GPU infrastructure for a credible SIH prototype. Engineering simplicity is more valuable than fancy distributed training.

Data-to-pipeline mapping

Pipeline stage	Preferred source	Backup	Output
SAR detection	Sentinel-1 CDSE	archived Sentinel-1 benchmark	calibrated VV/VH raster
Optical check	Sentinel-2	Landsat	optical evidence

Pipeline stage	Preferred source	Backup	Output
Large-area context	MODIS	—	low-res contextual layer
False-positive environmental gate	ECMWF wind	ERA5	wind u, v
Hindcast	CMEMS SMOC	HYCOM	surface u, v
Cross-validation currents	OSCAR	NOAA ERDDAP	alternate current field
AIS training/demo	MarineCadastre	synthetic	historical vessel positions
Near-real-time AIS	MarineTraffic	AISHub where available	current vessel tracks
ML training	Zenodo oil dataset	PANGAEA	model weights
Look-alike benchmark	PANGAEA + Zenodo III	—	false-alarm metrics
Drift engine	OpenDrift	GNOME	particle trajectories
Evidence	all above	—	provenance package

Sentinel-1 download example

CDSE's OData endpoint supports product discovery and authenticated download. The official service uses OData/STAC plus CDSE authentication; credentials should come from environment variables, not source code. ³¹

```

from __future__ import annotations

import os
from pathlib import Path

import requests

CATALOG = "https://catalogue.dataspace.copernicus.eu/odata/v1/Products"
TOKEN_URL = (
    "https://identity.dataspace.copernicus.eu/auth/realms/CDSE/"
    "protocol/openid-connect/token"
)
DOWNLOAD_BASE = "https://download.dataspace.copernicus.eu/odata/v1"

def get_access_token() -> str:

```

```

username = os.environ["CDSE_USERNAME"]
password = os.environ["CDSE_PASSWORD"]

response = requests.post(
    TOKEN_URL,
    data={
        "client_id": "cdse-public",
        "username": username,
        "password": password,
        "grant_type": "password",
    },
    timeout=30,
)
response.raise_for_status()
return response.json()["access_token"]

def find_sentinel1() -> dict:
    # Example point in Arabian Sea; replace AOI + dates for your incident.
    filter_expr = (
        "Collection/Name eq 'SENTINEL-1' and "
        "OData.CSC.Intersects("
        "area=geography'SRID=4326;POINT(72.0 18.0)') and "
        "ContentDate/Start gt 2026-08-01T00:00:00.000Z and "
        "ContentDate/Start lt 2026-08-03T00:00:00.000Z"
    )

    response = requests.get(
        CATALOG,
        params={
            "$filter": filter_expr,
            "$top": 20,
            "$orderby": "ContentDate/Start desc",
        },
        timeout=60,
    )
    response.raise_for_status()

    products = response.json()["value"]

    # In production additionally filter for GRD/IW and desired polarization.
    grd = [p for p in products if "_GRD" in p["Name"]]
    if not grd:
        raise RuntimeError("No matching Sentinel-1 GRD product found.")

    return grd[0]

```

```

def download_product(product: dict, destination: Path) -> None:
    token = get_access_token()

    url = f"{DOWNLOAD_BASE}/Products({product['Id']})/$value"

    with requests.get(
        url,
        headers={"Authorization": f"Bearer {token}"},
        stream=True,
        timeout=300,
    ) as response:
        response.raise_for_status()

        with destination.open("wb") as fh:
            for chunk in response.iter_content(chunk_size=1024 * 1024):
                if chunk:
                    fh.write(chunk)

product = find_sentinel1()
print(product["Name"])

# Uncomment after configuring CDSE credentials:
# download_product(product, Path("sentinel1_product.zip"))

```

For production, prefer exact GRD product-type, polarization and AOI geometry filters rather than the simplified name check. CDSE also supports STAC and S3, which can become more efficient for large data volumes. ³²

AIS API example

AISHub's documented endpoint accepts geographic bounds and can return JSON, XML or CSV. It requires a participating/contributing account and should not be treated as guaranteed Indian coverage. ³³

```

from __future__ import annotations

import os

import requests

url = "https://data.aishub.net/ws.php"

params = {
    "username": os.environ["AISHUB_USERNAME"],
    "format": 1,
}

```



```

    "output": "json",
    "compress": 0,
    # Example Arabian Sea box
    "latmin": 17.0,
    "latmax": 20.0,
    "lonmin": 70.0,
    "lonmax": 74.0,
}

response = requests.get(url, params=params, timeout=60)
response.raise_for_status()

ais_records = response.json()
print(type(ais_records))

```

For SIH testing, NTRO's PS itself allows synthetic AIS when real data is unavailable. That is actually useful: tum controlled benchmark bana sakte ho jahan true source known ho.

Ocean-current download example

Copernicus Marine's current Global Analysis and Forecast service exposes hourly merged surface-current data and is a strong operational choice for the prototype. 34

```

# pip install copernicusmarine

import copernicusmarine

copernicusmarine.subset(
    dataset_id="cmems_mod_glo_phy_anfc_merged-uv_PT1H-i",

    # Example Arabian Sea incident box
    minimum_longitude=70.0,
    maximum_longitude=74.0,
    minimum_latitude=17.0,
    maximum_latitude=20.0,

    start_datetime="2026-08-01T00:00:00",
    end_datetime="2026-08-03T23:00:00",

    # Check the current dataset metadata before production use.
    variables=[
        "uo",
        "vo",
        "utide",
        "vtide",
    ]
)

```

```

        "ustokes",
        "vstokes",
        "utotal",
        "vtotal",
    ],

    output_filename="arabian_sea_surface_currents.nc",
)

```

The current product documentation identifies the merged surface-current dataset as hourly and includes general circulation, tides and Stokes components. ³

Eight-week build plan

Week	Main milestone	Concrete deliverable	Owners
W1	Data foundation	3–5 Sentinel-1 scenes, benchmark datasets, AIS schema, CMEMS/ECMWF API working	RS + Ocean + AIS
W2	SAR preprocessing	reproducible SNAP/GDAL pipeline + training chips	RS + ML
W3	Detection baseline	U-Net + SegFormer experiment; IoU/precision/recall dashboard	ML + RS
W4	False positives + UI	look-alike classifier, map showing SAR/mask/confidence	ML + Frontend
W5	Hindcast	OpenDrift integrated with CMEMS and wind; 50/90% origin contours	Ocean
W6	AIS attribution	trajectory cleaning, candidate retrieval, explainable ranking, unknown class	AIS + Ocean
W7	Full integration	one-click case analysis + PDF/GeoJSON + provenance	Backend + Frontend + MLOps
W8	Validation & jury	historical case, synthetic source tests, ablation results, polished demo/pitch	Entire team

Non-negotiable milestone: end of W5 tak **image** → **segmentation** → **hindcast** working hona chahiye. W7 tak **image** → **ranked vessel report**.

Risk, Validation and Six-Member Execution Plan

Risk register

Risk	Probability	Impact	Mitigation
SAR look-alikes	High	High	dedicated look-alike training + wind/texture/polarization features
Optical cloud cover	High	Medium	optical ko optional evidence rakho; SAR primary
Sentinel scene huge / slow preprocessing	High	Medium	AOI crop, COG/RTC, cached preprocessing
Dataset domain shift	High	High	event-level test split, geographic holdout
Spill age unknown	High	High	age range/ensemble, not single age
Ocean-current error	High	High	CMEMS ensemble/sensitivity; HYCOM/OSCAR cross-check
AIS gaps	High	High	uncertainty penalty + unknown-source hypothesis
AIS spoofing / wrong static fields	Medium/High	High	kinematic anomaly checks; do not trust ship type alone
Vessel without AIS	Medium	Critical	unknown/non-AIS candidate
False legal attribution	Medium	Critical	investigative-score wording, confidence, provenance, analyst review
Paid AIS unavailable	High for students	Medium	MarineCadastre/synthetic data for demo
No real confirmed culprit labels	High	High	synthetic source injection + documented historic cases
GPU limits	Medium	Low	smaller SegFormer/U-Net; tile inference
API failure during jury	Medium	High	cache demo case completely offline

IMO's warnings around incomplete AIS and its use as only one input justify treating AIS gaps and unknown vessels as first-class design risks. ²

Validation strategy

Oil detector

Report on held-out datasets:

- IoU
- Dice/F1
- precision
- recall
- look-alike false-positive rate
- PR-AUC
- inference time

Use Zenodo Part III and PANGAEA look-alike benchmark independently. 35

Drift model

Known origin x_{true} ke against:

$$E_{origin} = distance(x_{pred}, x_{true})$$

Also:

- temporal origin error
- true origin inside 50% contour?
- true origin inside 90% contour?
- forward position error after N hours

GNOME vs OpenDrift outputs compare karo to catch obvious forcing/configuration bugs. Both are established trajectory-modelling frameworks. 36

AIS attribution

Synthetic benchmark strongest hai.

Example:

Create 30 vessels.

Secretly Vessel 17 ko source choose karo.

Its position/time se 500 particles **forward** OpenDrift mein run karo.

Resulting slick ko observation treat karo.

Now original source hide kar do.

Your actual pipeline:

observed slick → hindcast → candidates → ranking

Then test:

$$Rank@1 = \frac{\text{cases where true vessel is rank 1}}{\text{total cases}}$$

Also measure:

- Rank@3
- Mean Reciprocal Rank
- score calibration
- “unknown source” detection
- performance when 10%, 30%, 50% AIS points removed

This is exceptionally strong jury validation because tum **ground truth control karte ho**.

Adversarial tests

Deliberately test:

AIS dropout: true source ke 20 minutes AIS remove.

Distractor: another vessel ko source se closer place karo but wrong time.

Direction trap: vessel nearby but opposite heading.

Low wind look-alike: non-oil dark patch.

Current error: forcing $\pm 10\text{--}20\%$ perturb.

Spill age uncertainty: true event ko plausible age range ke edge par place karo.

No vessel case: all candidate vessels remove karo; correct output should be “unknown”.

This proves your system ranking logic samajhta hai rather than nearest-ship heuristic.

Jury demonstration sequence

A powerful 5-minute demonstration:

First minute: Sentinel-1 raw scene kholo.

“Dark region visible hai, but every dark region oil nahi hota.”

Run AI.

"Oil probability 91%; look-alike risk low; area 8.4 km²."

Second minute: Hindcast button.

Hundreds of particles reverse move.

"Single point claim karne ke instead, we produce 50% and 90% source regions."

Third minute: AIS layer.

10–15 tracks appear.

System automatically filters:

"Only four vessels intersect the plausible space-time origin."

Fourth minute: Explain rank #1.

Vessel A	
Spatial match	92
Temporal match	88
Direction match	73
Origin overlap	90
AIS quality	97
Behaviour anomaly	42
Overall investigative compatibility: 0.79	
90% interval: 0.63–0.88	

Then click Vessel B and show why it is lower.

Fifth minute: show:

Unknown source: 0.18

and export forensic PDF.

Final pitch:

"We don't simply detect oil or choose the nearest vessel. We propagate uncertainty from the satellite image, through physical ocean drift, into explainable vessel attribution, while retaining the possibility that no observed AIS vessel caused the spill."

That is much stronger technically and ethically than a normal AI dashboard.

Six-member team split

Member	Primary role	Core responsibility	Secondary
Member A	ML/CV Lead	U-Net, SegFormer, evaluation	model serving
Member B	Remote Sensing Engineer	Sentinel/SNAP/GDAL, SAR features	dataset QA
Member C	Ocean Modelling Lead	OpenDrift, CMEMS, ECMWF	uncertainty ensemble
Member D	AIS + Backend Engineer	trajectories, scoring, FastAPI	PostGIS
Member E	Frontend/GIS Engineer	MapLibre/deck.gl, timeline, UX	PDF visualisation
Member F	Integration/MLOps/Validation	Docker, provenance, tests, reports	pitch/research

Week-by-week ownership

Weeks 1-2: A+B jointly own SAR dataset/preprocessing. C proves CMEMS/OpenDrift can run independently. D defines canonical AIS schema and synthetic traffic generator. E makes map shell. F establishes Git/Docker/test structure.

Week 3: A owns first validated model. B performs SAR feature extraction/quality checks. C builds reproducible forward/backward drift notebook. D builds AIS track interpolation. E visualizes masks/tracks. F freezes benchmark protocol.

Week 4: A+B focus look-alikes. D exposes inference APIs. E produces polished case UI. C adds environmental layers. F runs domain-shift tests.

Week 5: C becomes critical-path owner: image polygon → OpenDrift → origin probability envelope. E visualizes particle cloud and contours. A/B support geometry/age inputs.

Week 6: D owns candidate retrieval/scoring. C provides probabilistic origin data. A helps calibration. E builds explanation panel. F injects adversarial AIS cases.

Week 7: F owns integration freeze. D connects APIs, E completes reports, A/C benchmark runtime, B validates georeferencing.

Week 8: Nobody adds large new features. Team only fixes bugs, caches offline demo cases, validates metrics, prepares jury questions, records backup demo and finalizes evidence report.

What to cut when schedule slips

Cut in this order:

1. automated spill age estimation
2. Landsat/MODIS fusion
3. AIS spoof classifier
4. multiple segmentation ensembles
5. live MarineTraffic integration
6. multi-provider currents
7. fully automated incident monitoring

Never cut:

- oil vs look-alike confidence
- hindcast
- uncertainty envelope
- AIS ranking
- unknown-source option
- explainability
- provenance report

Those six items are the product's actual moat.

Final recommendation

The strongest final product name could be something like:

SAMUDRA-FI: Satellite-AIS Multi-modal Uncertainty-Driven Response & Attribution — Forensic Intelligence

The architecture should be presented as **decision support**, not automated guilt determination.

The central jury claim should be:

“Existing tools demonstrate that satellite oil detection, drift modelling and AIS are individually viable. Our contribution is an open, reproducible, uncertainty-aware forensic fusion pipeline that carries uncertainty from detection to source attribution and can explicitly conclude that none of the observed AIS vessels is sufficiently supported by evidence.”

CleanSeaNet proves the overall operational concept is real; KSAT proves commercial SAR-AIS fusion can run near real time; OpenDrift/GNOME provide proven trajectory foundations; Copernicus supplies strong free satellite/ocean data; and new open Sentinel-1 datasets make serious oil-vs-look-alike benchmarking possible. ³⁷

From an SIH execution perspective, that combination makes **PS 26143 technically ambitious but realistically prototype-able**—provided the team treats **vessel attribution and uncertainty as the core problem**, rather than stopping after a visually attractive oil-spill segmentation model.

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